Chapter 1

Introduction to Blockchain

Blockchain in India

- **Tech Mahindra** announced it was teaming up with Netherlands-based
  - blockchain application incubator Quantoz to provide secure digital payments.

- **Tata Consultancy Services** has also launched a multi-brand consumer loyalty
  - platform on R3’s enterprise blockchain Corda

- **India's defense minister** also stressed the potential use cases of blockchain in the defense industry .

Blockchain in India

- The Indian government has already built the Distributed Center of Excellence in Blockchain Technology.

- A piloted project on a blockchain system **for property registration** at Shamshabad District, Telangana State, https://www.youtube.com/watch?v=0qpeWxJPh2g

- **Other ongoing projects include**
  - **authentication of academic certificates with a proof-of-existence framework**
  - **Vehicle life cycle**
  - **Hotel registry management**

Blockchain

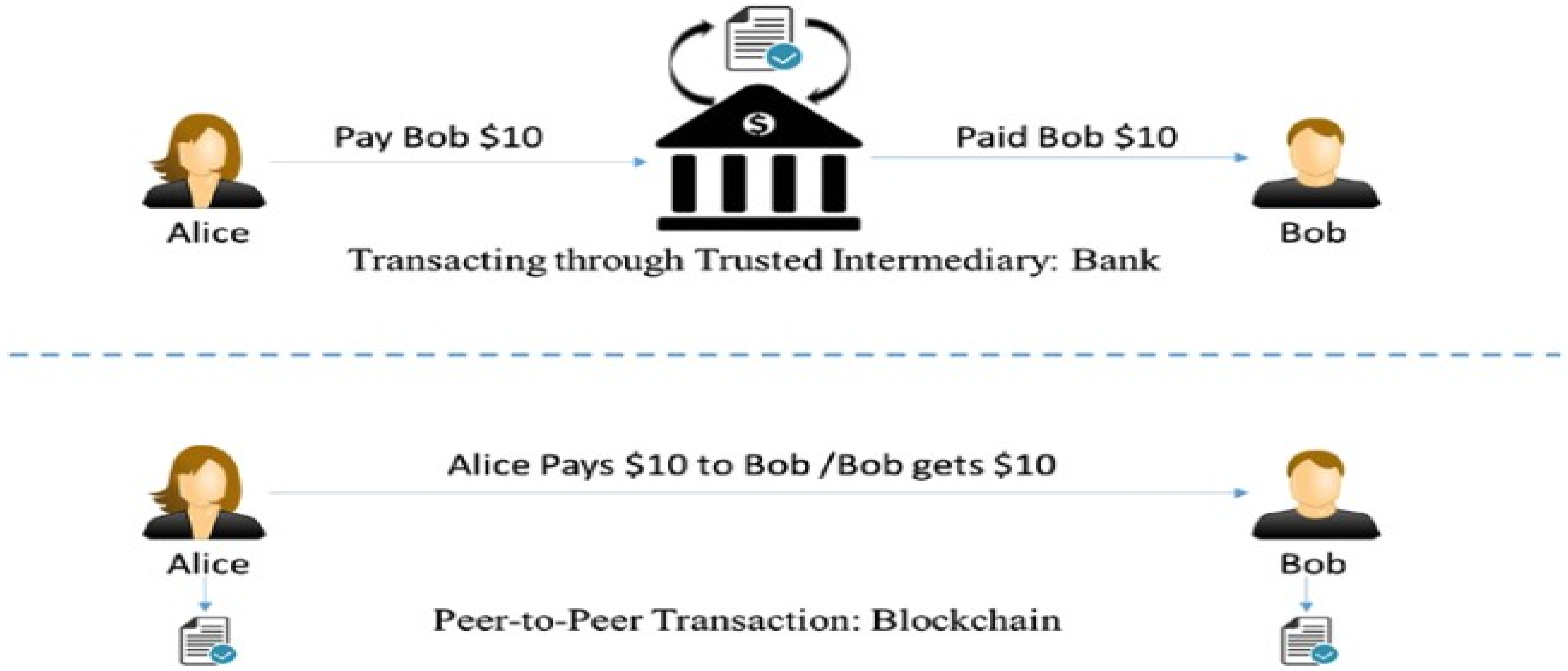


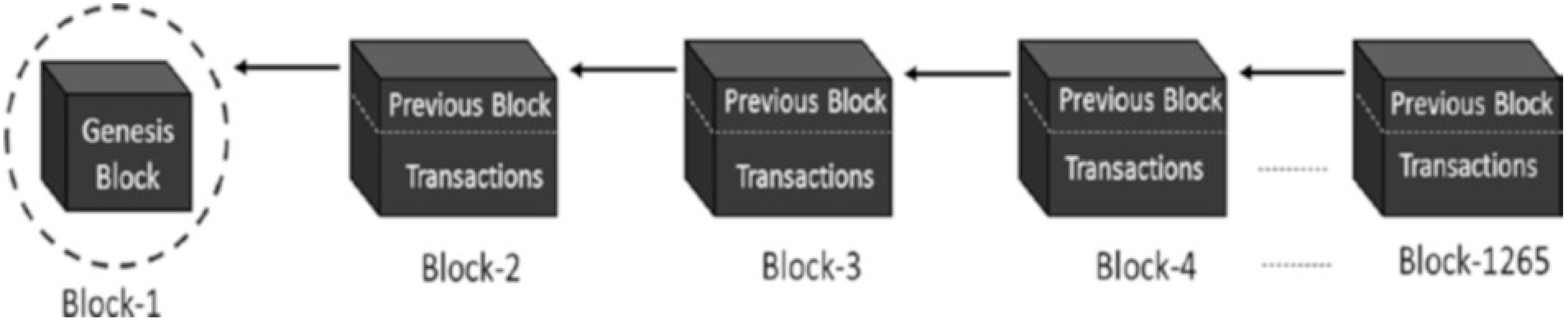
Figure 1-1. *Transaction through an intermediary vs. peer-to-peer transaction*

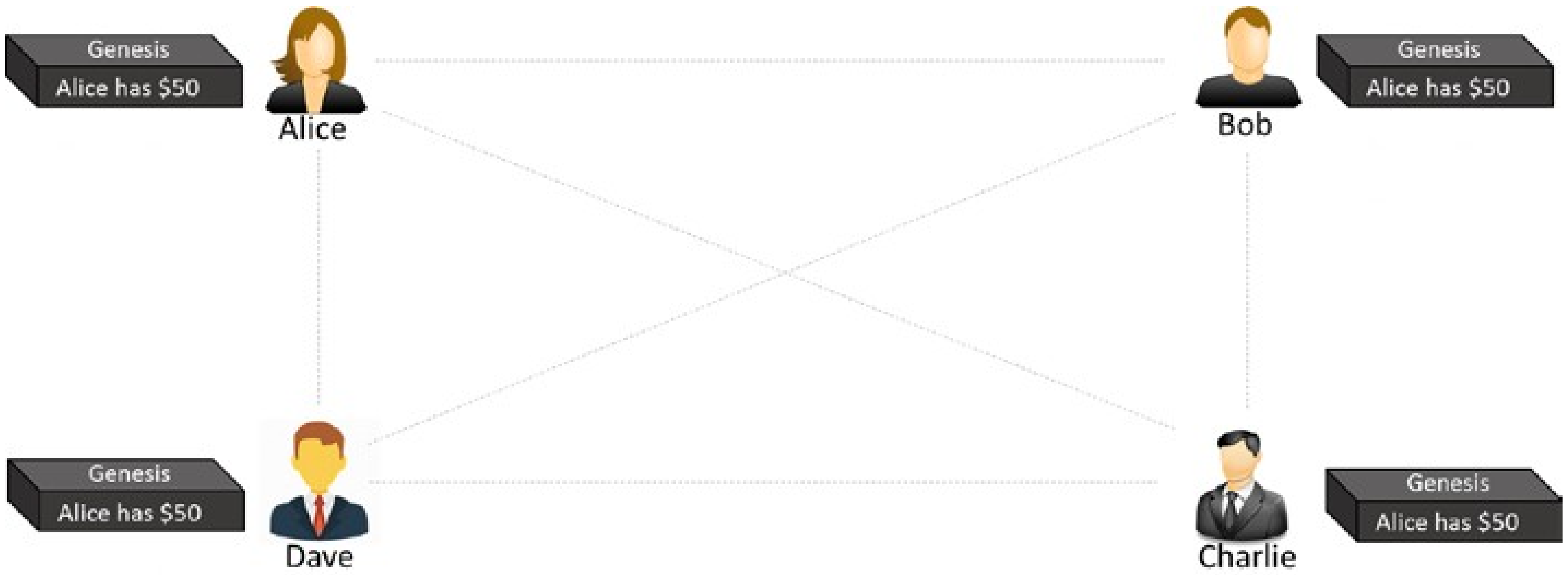
Blockchain

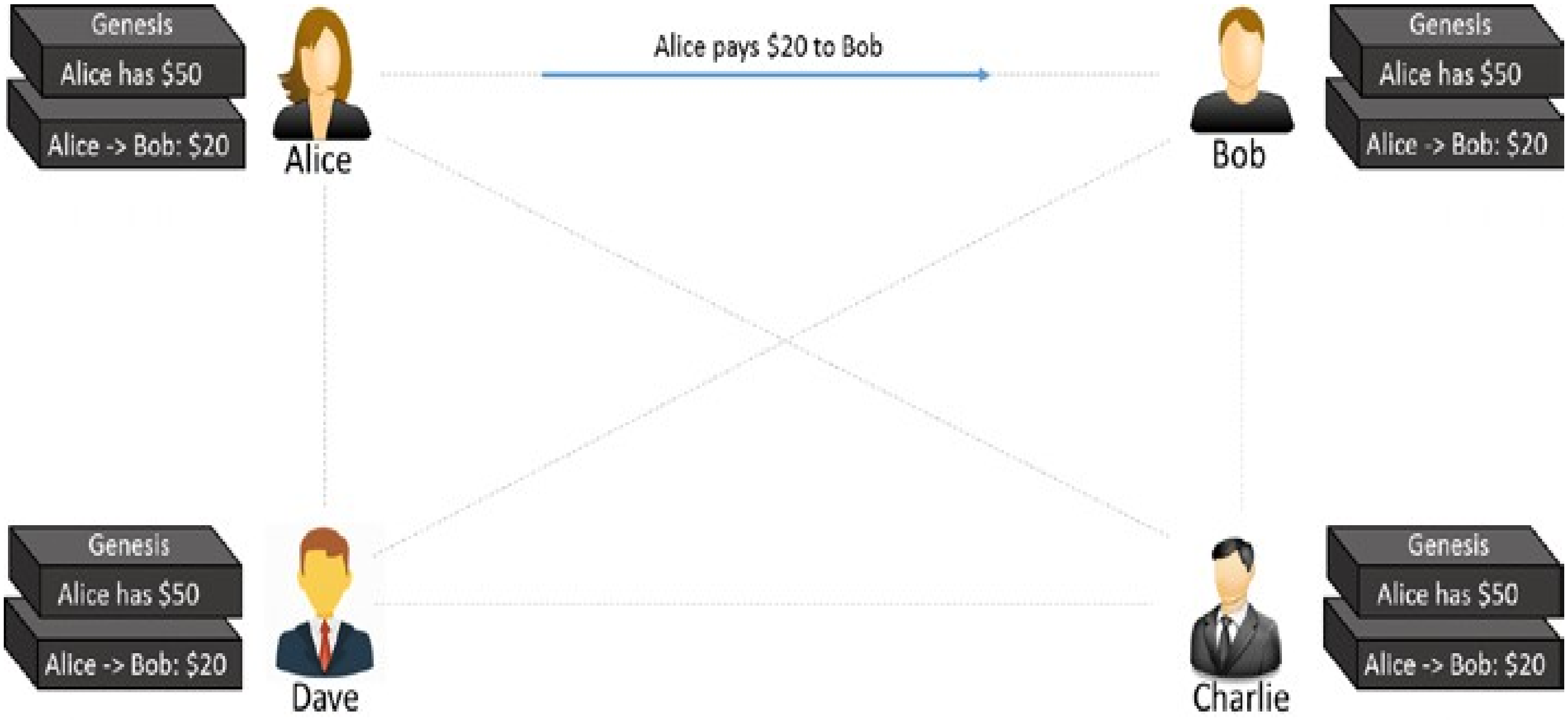
- Peer-to-peer system of transacting values
  - With no trusted third parties in between
  - Shared, decentralized, and open ledger of transactions
  - Ledger database is an append-only
  - Every entry is a permanent entry
  - Any new entry on it gets reflected on all copies of the databases hosted on different nodes.

BlockChain Basics

- BlockChain is a philosophy
  - It is a list of blocks
  - A distributed collection of immutable data or code
  - Linked and secured with some **crypto-hash(Fingerprint)**
  - Publicly available over the INTERNET
- Each child block has a crypto-hash of its parent block.


Genesis Block
Block-1
Previous Block
Transactions
Block-2
Previous Block
Transactions
Block-3
Previous Block
Transactions
Block-4
Previous Block
Transactions
Block-1265


Genesis
Alice has $50
Alice
Genesis
Alice has $50
Bob
Genesis
Alice has $50
Dave
Genesis
Alice has $50
Charlie


Genesis
Alice has $50
Alice -> Bob: $20
Alice
Alice pays $20 to Bob
Bob
Genesis
Alice has $50
Alice -> Bob: $20
Genesis
Alice has $50
Alice -> Bob: $20
Dave
Charlie
Genesis
Alice has $50
Alice -> Bob: $20


Genesis
Alice has $50
Alice -> Bob: $20
Bob-> Charlie: $10
Alice
Alice pays $20 to Bob
Bob
Genesis
Alice has $50
Alice -> Bob: $20
Bob-> Charlie: $10
Bob pays $10 to Charlie
Genesis
Alice has $50
Alice -> Bob: $20
Bob-> Charlie: $10
Dave
Charlie
Genesis
Alice has $50
Alice -> Bob: $20
Bob-> Charlie: $10

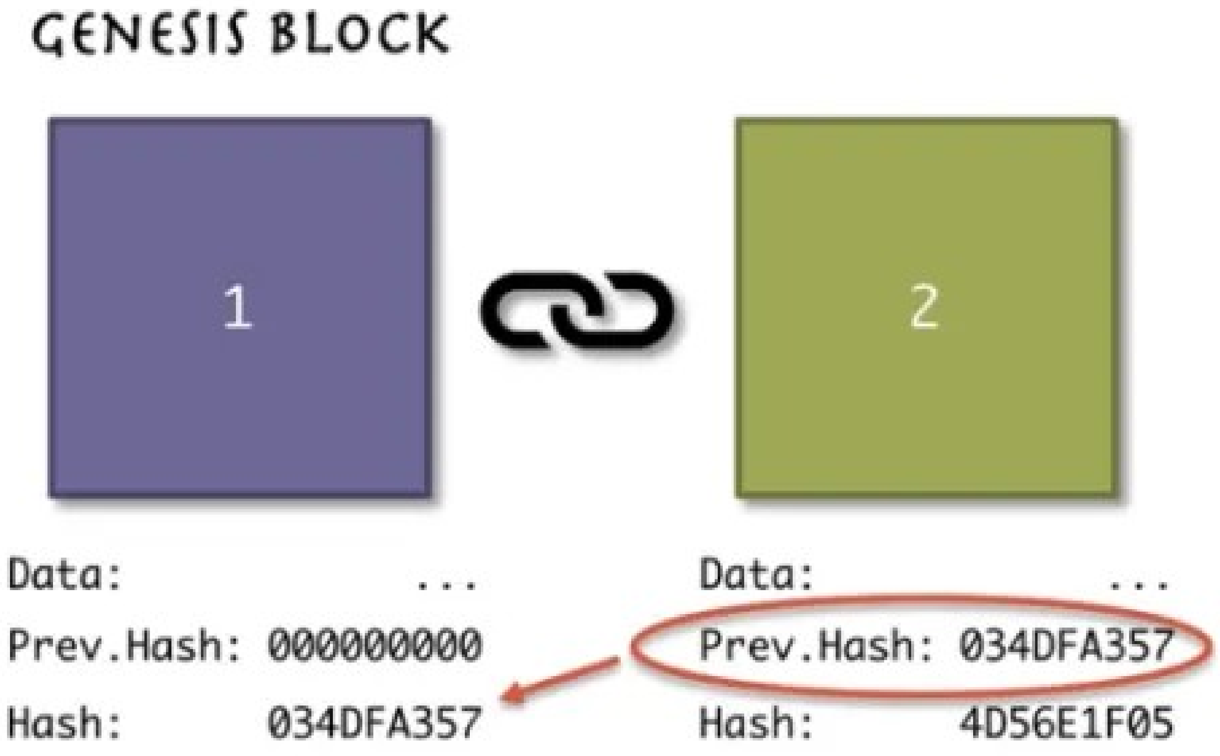
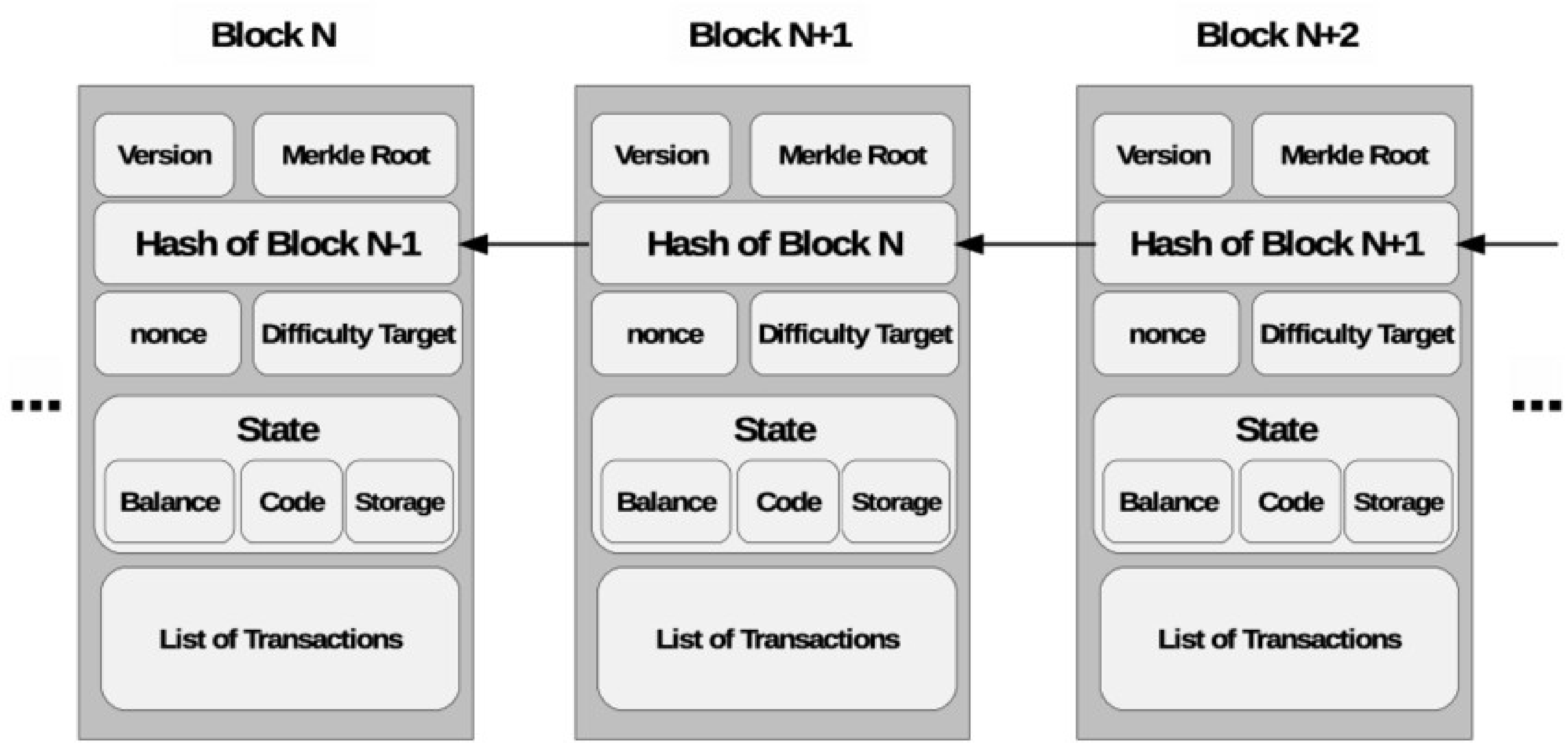


Figure-1: GENESIS_BLOCK is the first block in BlockChain, without having previous hash

There are two major parts in every block:

- The “**header**” part links back to the previous block in the chain with parent hash.

- The “**body content**” that has a
  - validated list of transactions
  - Their amounts
  - The addresses of the parties involved, and some more details.


Block N
Block N+1
Block N+2
Version
Merkle Root
Hash of Block N-1
Hash of Block N
Hash of Block N+1
nonce
Difficulty Target
State
Balance
Code
Storage
List of Transactions
...

More Terminologies

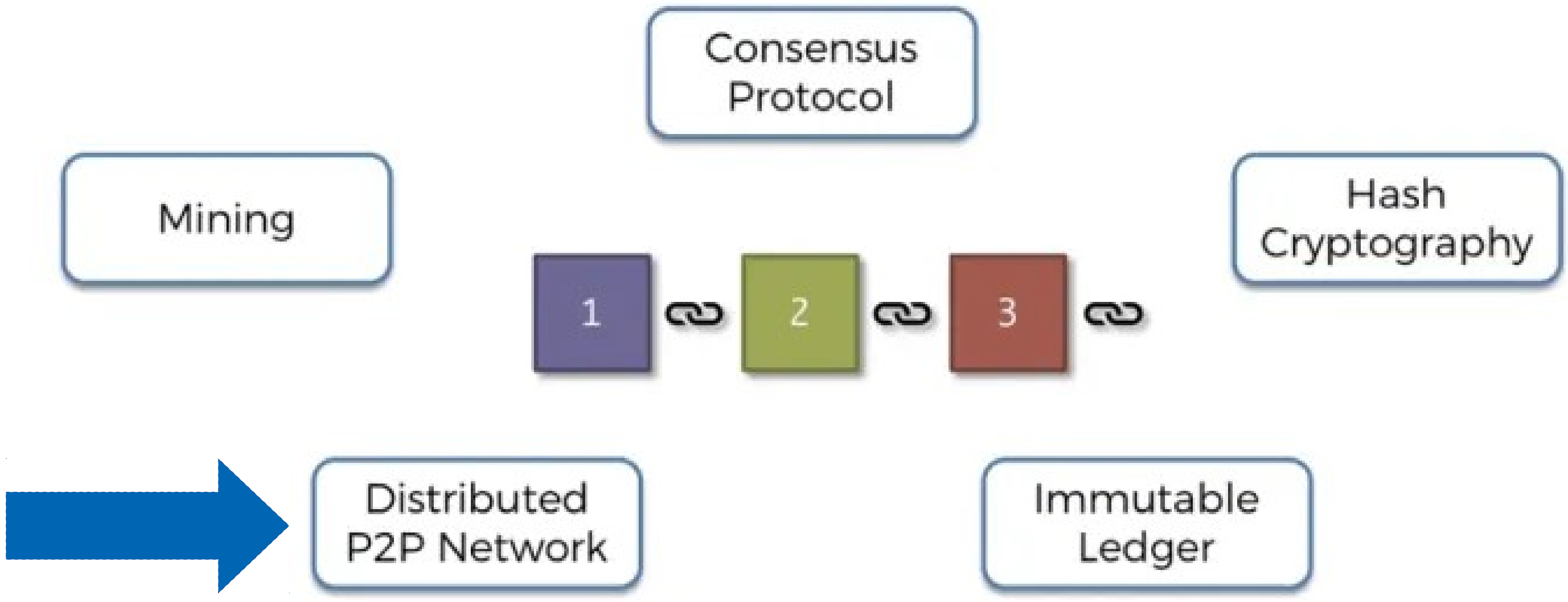
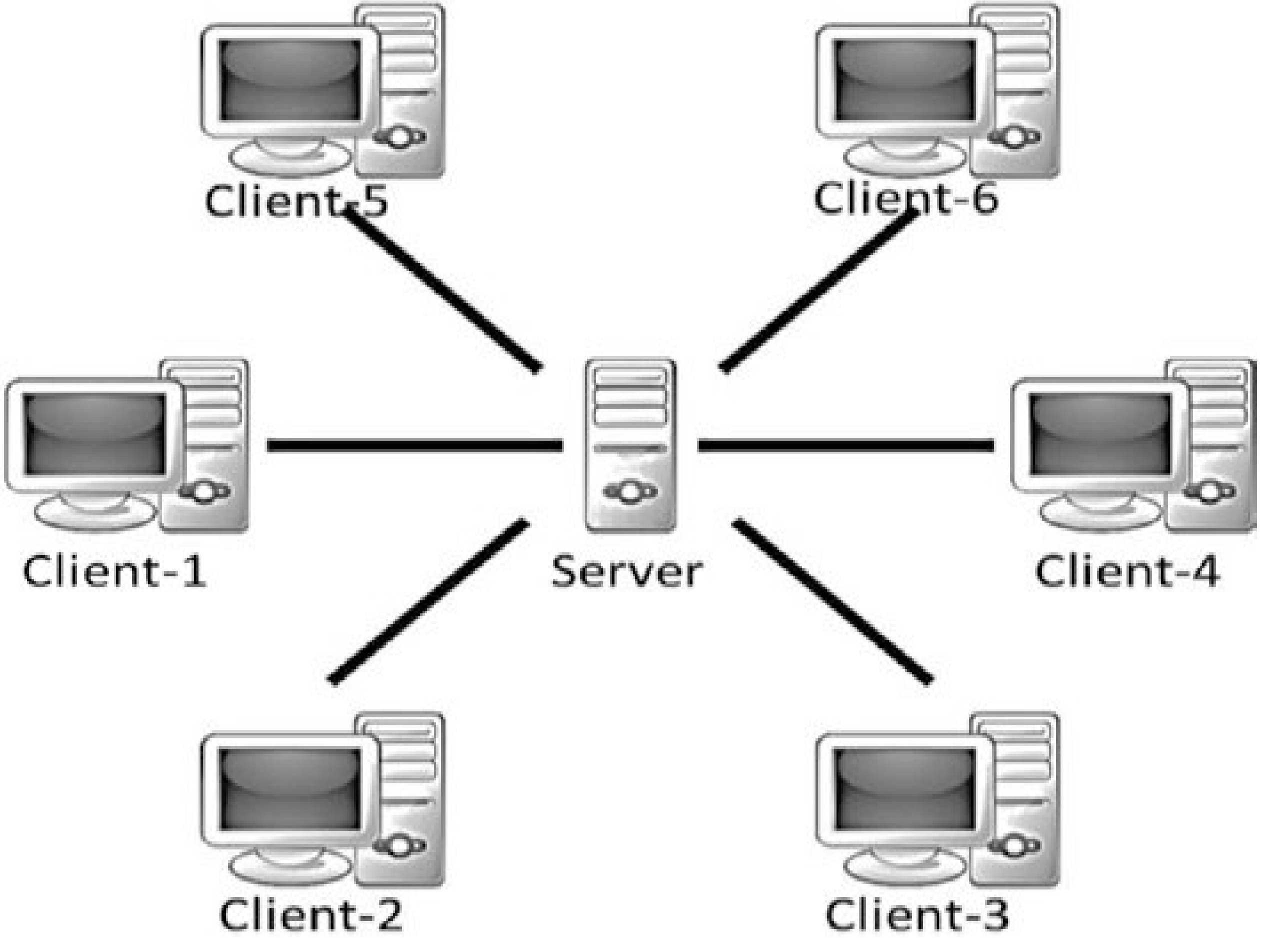


Figure-2: Fundamental building blocks for BlockChain technology

Centralized Systems
V/s
Decentralized Systems
V/s
Distributed Decentralized Systems

Centralized Systems



Centralized Systems

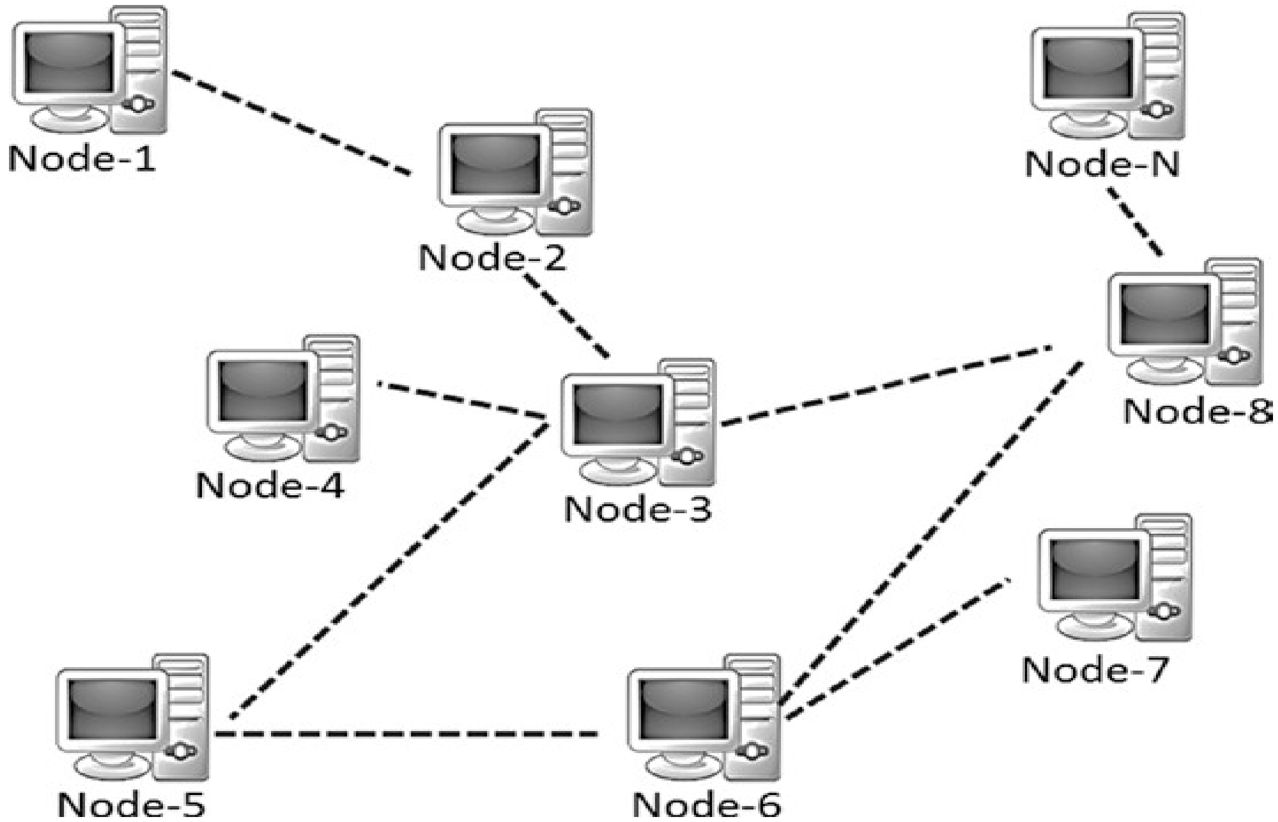
Advantages:

- Systems with a centralized control & all administrative authority.
- Such systems are <u>easy to design, maintain, impose trust, and govern</u>

Limitations:

- They have a central point of failure, so are less stable.
- They are more vulnerable to attack and hence less secured.
- Centralization of power can lead to unethical operations.
- Scalability is difficult most of the time.

Decentralized Systems



Decentralized Systems

Advantages:

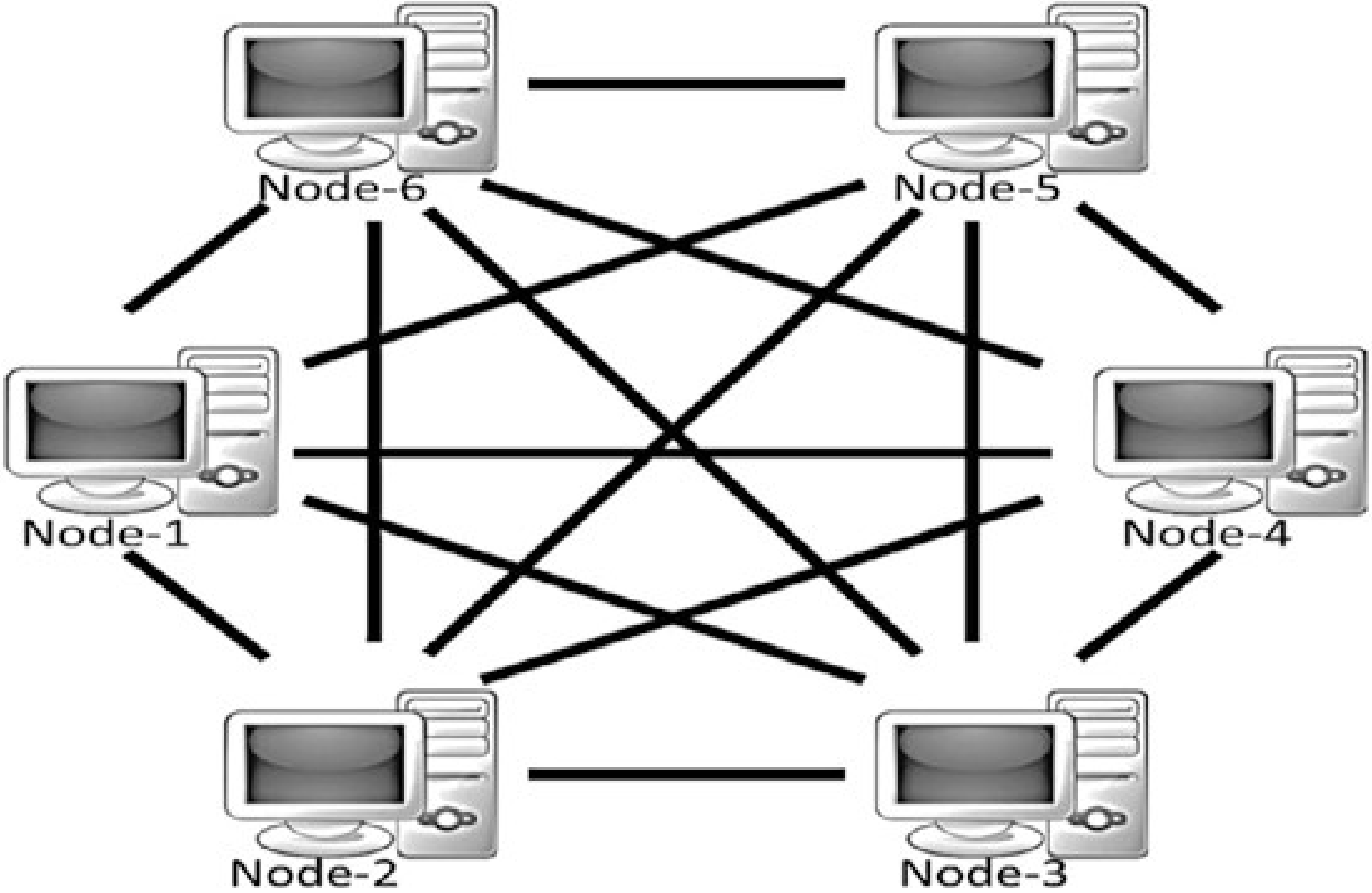
- They do not have a central point of failure, so <u>more stable and fault</u> tolerant
- Attack resistant, as no central point to easily attack and hence more secured
- Symmetric system with equal authority to all, so <u>less scope of unethical operations</u> and usually democratic in nature

Limitations:

- System does not have a centralized control and <u>every node has equal authority</u>.
- Such systems are **difficult** to design, maintain, govern, or impose trust.

Blockchain: A Distributed System

- It can also be **decentralized**.
- However, unlike common distributed systems, the task is not subdivided and delegated to nodes, as there is no master in blockchain.
- The contributing nodes do not work on a portion of the work,
- Rather, the interested nodes (or **the ones chosen at random**) perform the entire work.


Node-6
Node-5
Node-1
Node-4
Node-2
Node-3

Layers of Blockchain

- Blockchain is a **combination of:**
  - business principles & Economics
  - game theory & Cryptography
  - computer science engineering
- The layered approach in the TCP/IP stack is a standard to achieve an open system.
- Rather, In the blockchain, there are no agreed global standards.
- **A layered heterogeneous architecture is needed in future.**
- All these layers are present on all the nodes.

High-level, layered representation of blockchain

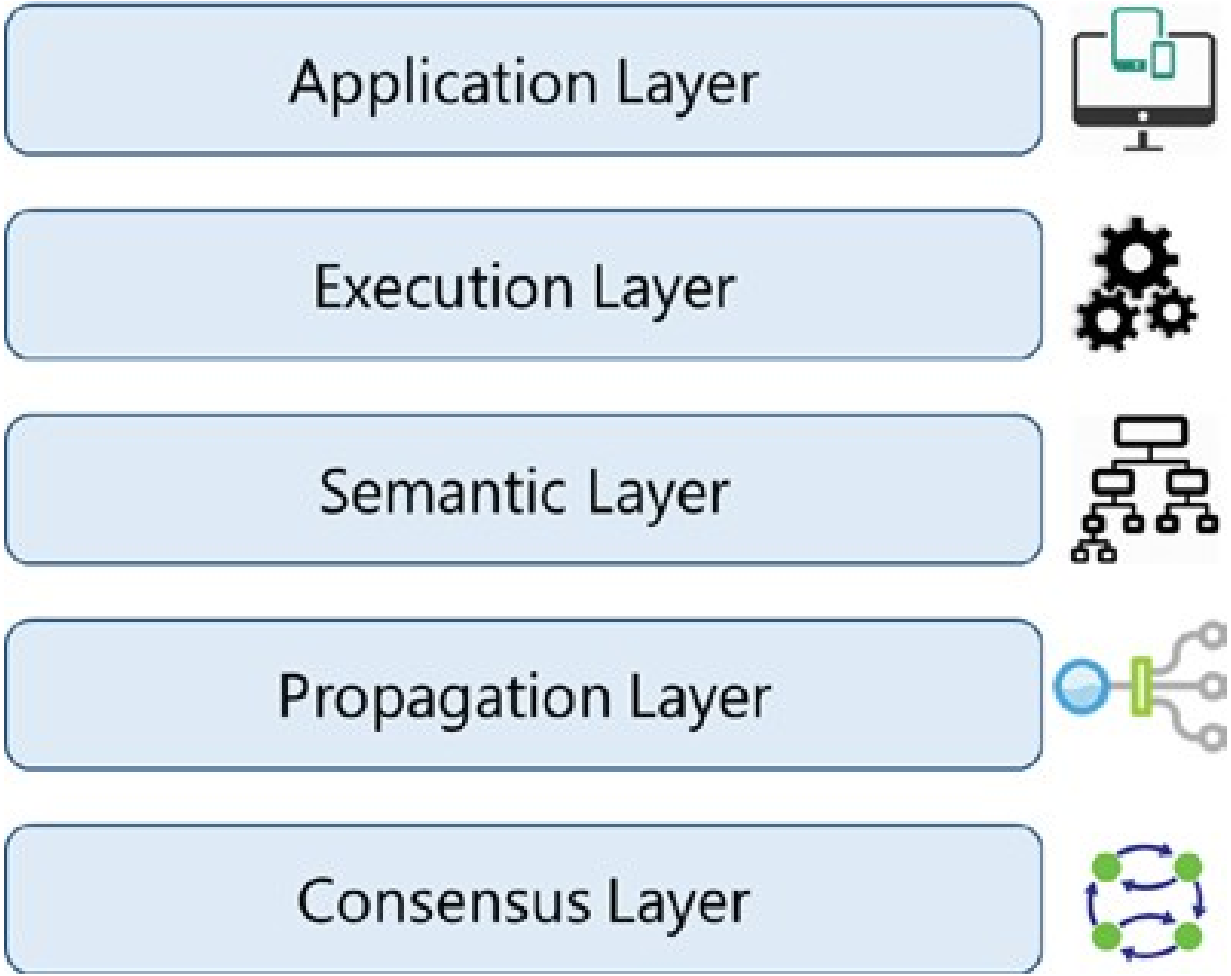
Coding level functionality: for front-end & Back-end

Execution of Single or set of Instruction by nodes

Validating the transaction

Allows the nodes to discover each other, talk and sync

Assuring Safety and security



Application Layer

- It is used to code up the desired functionalities
  - client-side programming / scripting / APIs
  - development frameworks
- **Off-chain networks is used**:
  - Transactions occur outside of the primary blockchain network
  - To ensure heavy lifting is done at the application layer
  - so core blockchain is light and effective
  - network traffic is less

Execution Layer

- Executions of instructions ordered by the Application Layer
- Instructions could be
  - <u>simple instructions</u>
  - <u>or a set of multiple instructions,</u> a ***smart contract***
- All the nodes execute the **<u>programs /scripts independently.</u>**
- e.g: Ensures transparency by recording every stage of a product’s journey on the blockchain
- https://www.youtube.com/watch?v=nbcJk4TZUP4

Execution Layer at various crypto-currencies

- **Bitcoins** uses simple scripts and allow few set of instructions.
- **Ethereum** allow <u>complex executions</u> that gets executed on its own **EVM**(Ethereum Virtual Machine).
- **Hyperledger** uses <u>smart contracts</u> using inside docker

  (list of many executable files) images

  – supports languages such as **Java and Go**.

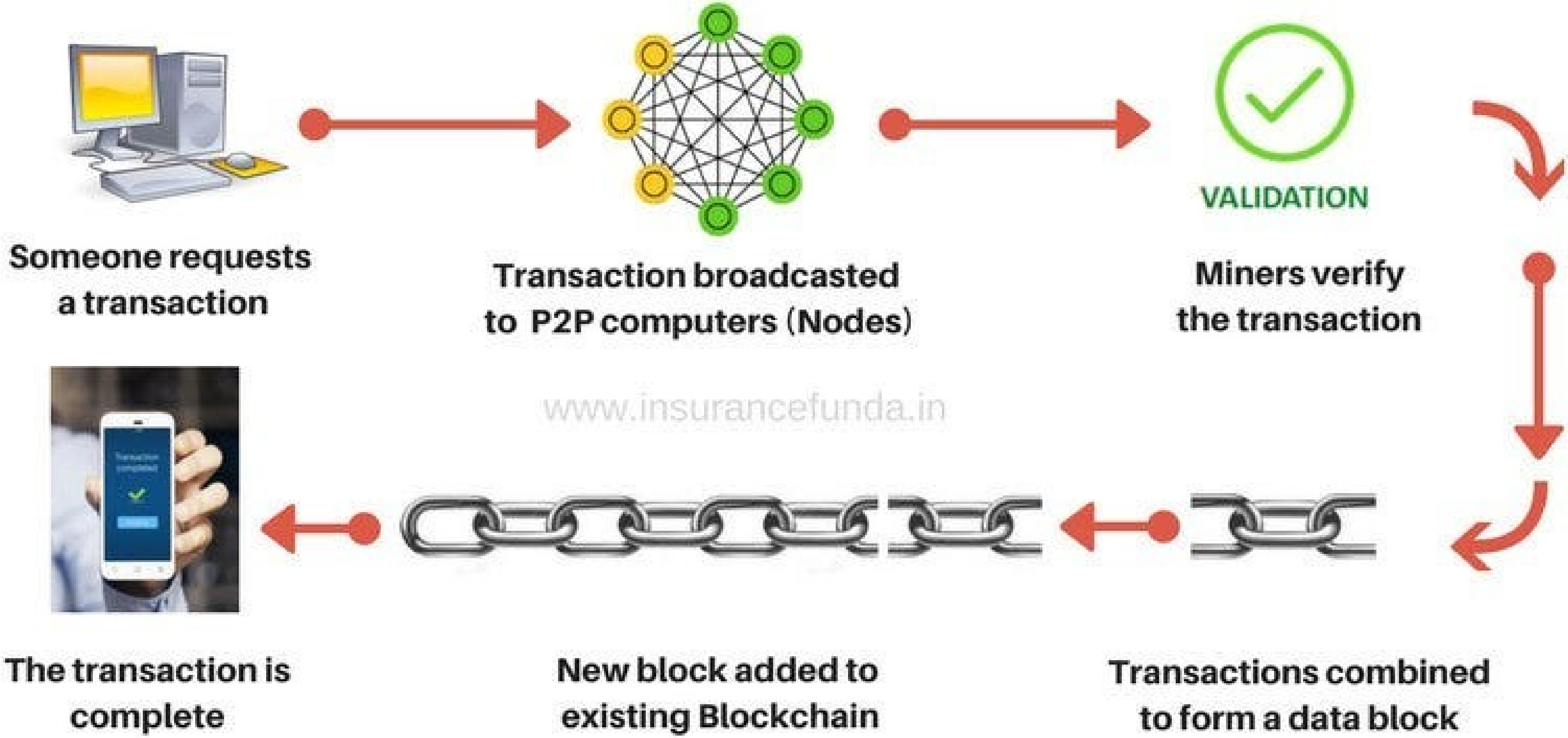
Semantic Layer

- Is a **logical layer**
- A transaction, whether valid or invalid, <u>gets validated in the Semantic Layer</u>.
- It will check:
  - legitimate transaction
  - double-spend attack
  - Is the node authorized to make this transaction
  - <u>Check one or more previous transactions</u>

Propagation Layer

- Peer-to-peer **communication layer, allows the nodes to sync**
- When a transaction is made, it gets broadcasted to the entire network.
- When a **node wants to propose a valid block**, it gets immediately propagated to the entire network as the latest block.
- **Latency issues for transaction or block:**
- Some propagations occur within seconds or after some time, depending on the
  - capacity of the nodes
  - network bandwidth

HOW BITCOIN TRANSACTION WORKS



Process of proposing a New Block

1. Transaction Collection

Unconfirmed Transactions: Miners gather unconfirmed transactions from the mempool (memory pool) – a holding area for transactions that have not yet been included in a block.

2. Transaction Verification: Before mining, miners check that the transactions are valid

Validation: Digital signatures are correct., The sender has enough balance to perform the transaction.

3. Mining the Block

Hashing the Block Header: Miners start trying to find a valid hash for the block header.

The block header is passed through a cryptographic hashing algorithm (typically SHA-256 in Bitcoin).

Process of proposing a New Block

4. Validating and Broadcasting the Block

The valid hash is a proof of work, indicating that the miner has expended computational resources to find it.

5. Block Addition to the Blockchain

If the block is **valid**, it is added to the blockchain, and all nodes update their ledger.

Consensus: If The network reaches consensus on the new block, and the blockchain is updated with the latest block.

6. Reward for the Miner

The miner who successfully mined the block is rewarded with:

Block reward: Newly minted cryptocurrency (e.g., 6.25 BTC in Bitcoin as of 2024)

Consensus(general agreement) Layer

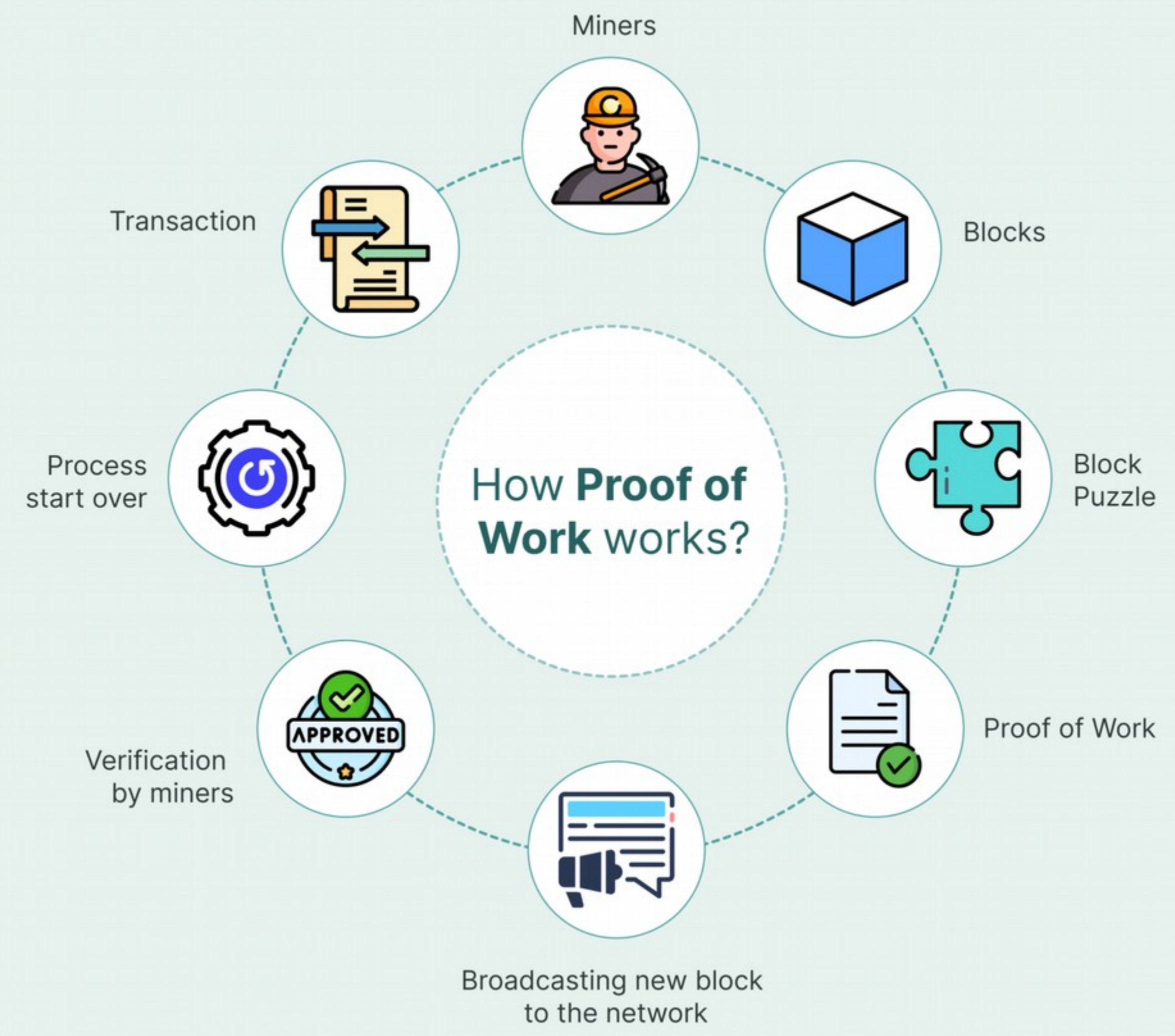
- It is a base layer assuring Safety and security
- Allow to get all the nodes to agree on one consistent state of the ledger.
- In Bitcoin or Ethereum,
  - it is used through “mining”
  - It use a **Proof of Work (PoW)** consensus mechanism to randomly select a node that can propose a block.

Consensus Layer

- If the PoW puzzle was solved properly
- <u>They add this block to their own copy of blockchain and build further on it.</u>
- There are many different variants of consensus protocols such as
  - **Proof of Work (PoW)**
  - **Proof of Stake (PoS)**
  - **deligated PoS (dPoS)**
  - **Practical Byzantine Fault Tolerance (PBFT)**
  - **etc**

Proof-of-Work

- <u>Original consensus algorithm</u>(mathematical Puzzle)
- **Miners compete against each other** to complete transactions on the network
- The right solution **they announces** it to the whole n/w
- Receiving a <u>crypto currency prize</u> provided by the protocol PoW.
- Miners use more and more powerful machines due to increase of complexity
- e.g: Bitcoin (BTC), Litecoin (LTC), Dogecoin (DOGE)

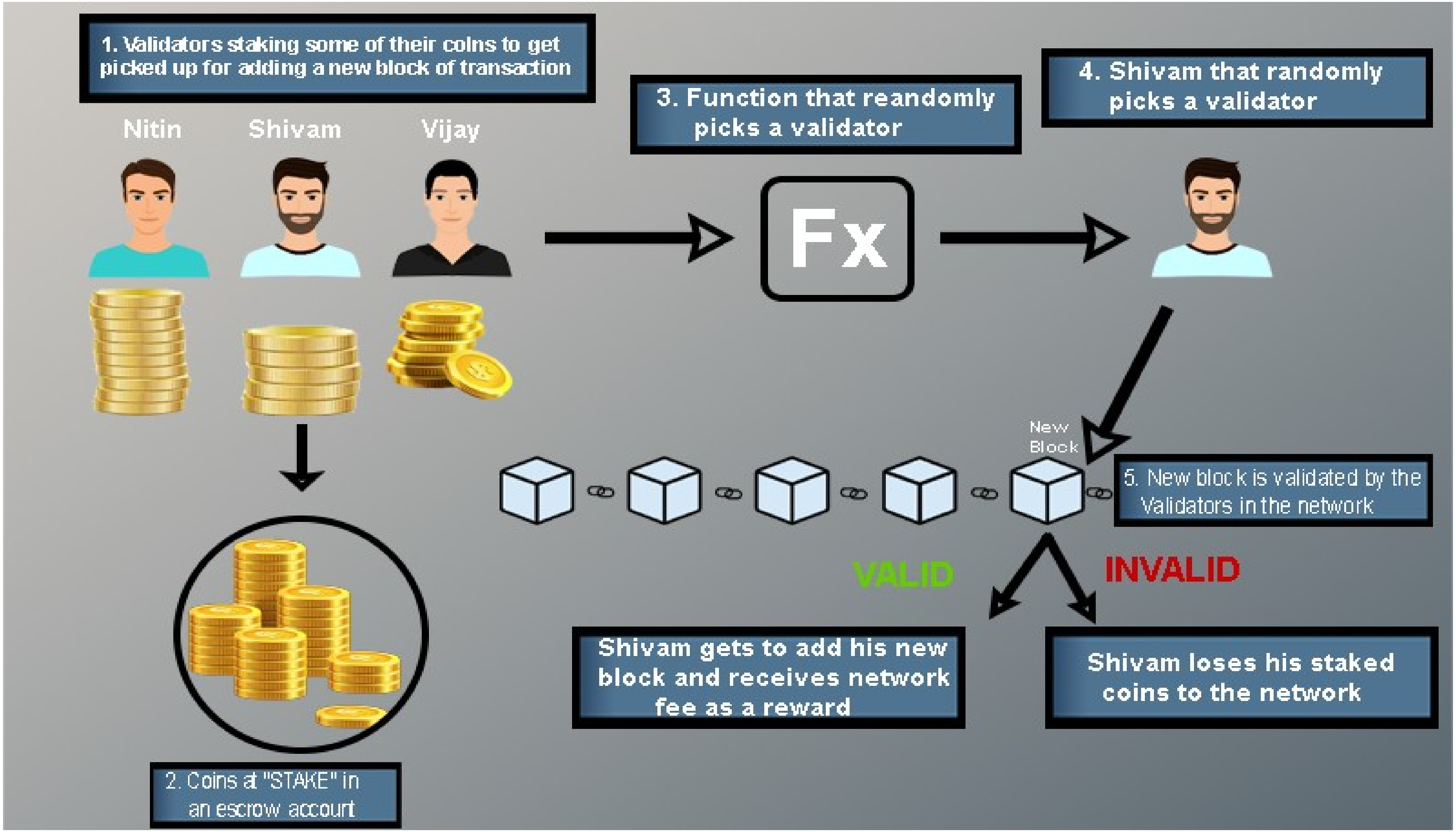

Miners
Blocks
Block Puzzle
Proof of Work
Broadcasting new block to the network
Verification by miners
Process start over
Transaction
How Proof of Work works?
APPROVED

Challenges to Proof-of-Work

- **51% Attack**
- A 51% attack occurs when a malicious actor controls more than 50% of the network's mining power.
- This would allow them to manipulate transactions, double spend, or block valid transactions.
- **Orphaned Blocks**
- If a miner mines an invalid block or a block that is not chosen by the majority
- The block is considered orphaned and the miner loses the block reward.

Proof-of-Stake

- Proof-of-stake (POS) was created as an alternative to proof-of-work (POW)
- While PoW mechanisms require miners to solve cryptographic puzzles,
- PoS mechanisms require validators to hold and stake tokens for the privilege of earning transaction fees.
- Under proof-of-stake (POS), validators are chosen based on the **number of staked coins they have.**
- e.g: Ethereum (ETH 2.0), Cardano (ADA), Solana (SOL).


1. Validators staking some of their coins to get picked up for adding a new block of transaction
Nitin
Shivam
Vijay
2. Coins at "STAKE" in an escrow account
3. Function that reandomly picks a validator
Fx
4. Shivam that randomly picks a validator
New Block
5. New block is validated by the Validators in the network
VALID
INVALID
Shivam gets to add his new block and receives network fee as a reward
Shivam loses his staked coins to the network

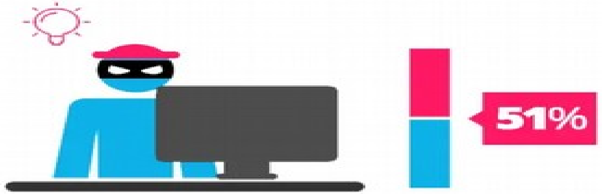
Process at Proof-of-Stake

- **Validators Locking Coins**: Users deposit a certain amount of cryptocurrency into the network.
- **Becoming a Validator**: If the stake meets the minimum requirement (e.g., 32 ETH for Ethereum 2.0), the user can validate transactions.
- **Earning Rewards**: Validators receive staking rewards for confirming transactions and securing the network.
- **Slashing** is a penalty mechanism in Proof of Stake (PoS) blockchains that punishes validators for misbehavior or failing to follow network rules.
- **When a validator is slashed**, they lose a portion of their staked cryptocurrency as a punishment.
- **The network itself** (through smart contracts or protocol rules) or other validators/nodes may also report malicious behavior.

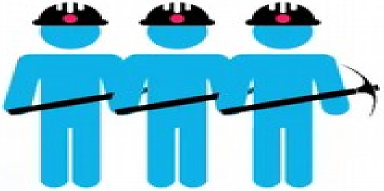
Proof of Work vs. Proof of Stake

Proof of Work

To add each block to the chain, miners must compete to solve a difficult puzzle using their computers processing power.



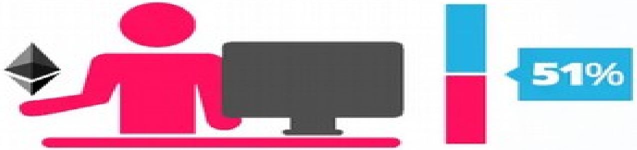
In order to add a malicious block, you'd have to have a computer more powerful than 51% of the network.



The first miner to solve the puzzle is given a reward for their work.

Proof of Stake

There is no competition as the block creator is chosen by an algorithm based on the user's stake.



In order to add a malicious block, you'd have to own 51% of all the cryptocurrency on the network.

There is no reward for making a block, so the block creator takes a transaction fee.

Feature	**Mining (PoW)**	**Staking (PoS)**
Process	Miners solve cryptographic puzzles to validate transactions.	Validators are selected based on the amount of cryptocurrency they stake.
Energy Consumption	High (requires powerful hardware).	Low (less powerful mining hardware needed)
Reward System	Miners earn block rewards and transaction fees.	Validators earn staking rewards and transaction fees.
Security	Secured by computational difficulty.	Secured by financial commitment (slashing penalty for malicious behavior).
Examples	Bitcoin (BTC), Dogecoin (DOGE), Litecoin (LTC).	Ethereum 2.0 (ETH), Cardano (ADA), Solana (SOL).

Software / Libraries for Blockchain

- **Decentralized Application (dApp) Platforms**
- Ethereum – Most popular blockchain for dApps.
- Binance Smart Chain (BSC) – Fast and cost-effective alternative to Ethereum.
- Solana – High-performance blockchain for dApps.
- Polkadot – Enables interoperability between blockchains.

Mining

- Mining is the process of verifying **bitcoin transactions** and storing them in a blockchain(ledger).
- It is a process similar to gold mining but instead, it is a complicated computer code to generate a secure cryptographic system.
- The bitcoin miner is the person who solves mathematical puzzles(also called proof of work) to validate the transaction.
- The miners need **to find the perfect solution hash** that matches and fits.
- Anyone with mining hardware and computing power can take part in this.
- Numerous miners take part simultaneously to solve the complex mathematical puzzle

Who can propose a Block?

- 1. Join a Mining Pool (Most Common Method)
  - Instead of mining solo, you join a mining pool (e.g., F2Pool, Slush Pool, Antpool).
  - The pool collectively mines blocks and shares rewards among members.
  - The pool operator is responsible for proposing blocks on behalf of the miners.
- 2. Cloud Mining Services
  - You rent mining power from a company like Genesis Mining, NiceHash, or Bitdeer.
  - The company operates the mining hardware and proposes blocks.
  - You earn a share of the mining rewards based on your contract.
- 3. Mining as a Service (MaaS)
  - Some specialized firms offer Mining as a Service, where they mine on your behalf and share profits with you.
  - Similar to cloud mining but with more transparency and customizability.

Mining

- Miners are using:
  - **Specialized mining hardware: “application-specific integrated circuits(ASICs).**
  - **A Bitcoin mining software to join the Blockchain network.**
  - **Powerful GPU (graphics processing unit).**
  - **Bitcoin Wallet: To be kept the bitcoin, after the user receives bitcoins as a reward for mining**
  - **The winning miner gets a block reward (e.g., 3.125 BTC in Bitcoin)**
  - **You cannot mine PoS coins (Ethereum 2.0, Cardano) because they use staking instead of mining**

What are smart contracts?

- Lines of code that are stored on a blockchain
- **Automatically execute** when predetermined terms and conditions are met.
- They are programs run by the people who developed them.
- In business collaborations, used to enforce some type of agreement

The structure of any smart contract includes the following attributes:

- The contract parties that have accepted the agreed conditions (for this, an electronic signature or multi-signatures, if there are several parties, is used);
- The environment in which the contract will be located (for example, Ethereum);
- The subject of the contract – namely, resources for exchange;
- Terms of the contract – a description of the mathematically confirmed conditions under which the contract will be considered as fulfilled.

A smart contract will store all of the listed funds until the set goal is reached. If the goal is not achieved, then the money is returned to investors.

Smart contracts in Supply Chain

- Smart Contract Triggers
  - Payment release on successful delivery confirmation
  - Retailer finalizes order → Payment automatically sent to distributor.
  - If GPS tracking shows unexpected route deviation, an alert is triggered for possible theft or fraud.
  - If stock level falls below a predefined threshold, the smart contract automatically places a new order with the supplier.
  - If GPS tracking shows unexpected route deviation, an alert is triggered for possible theft or fraud.

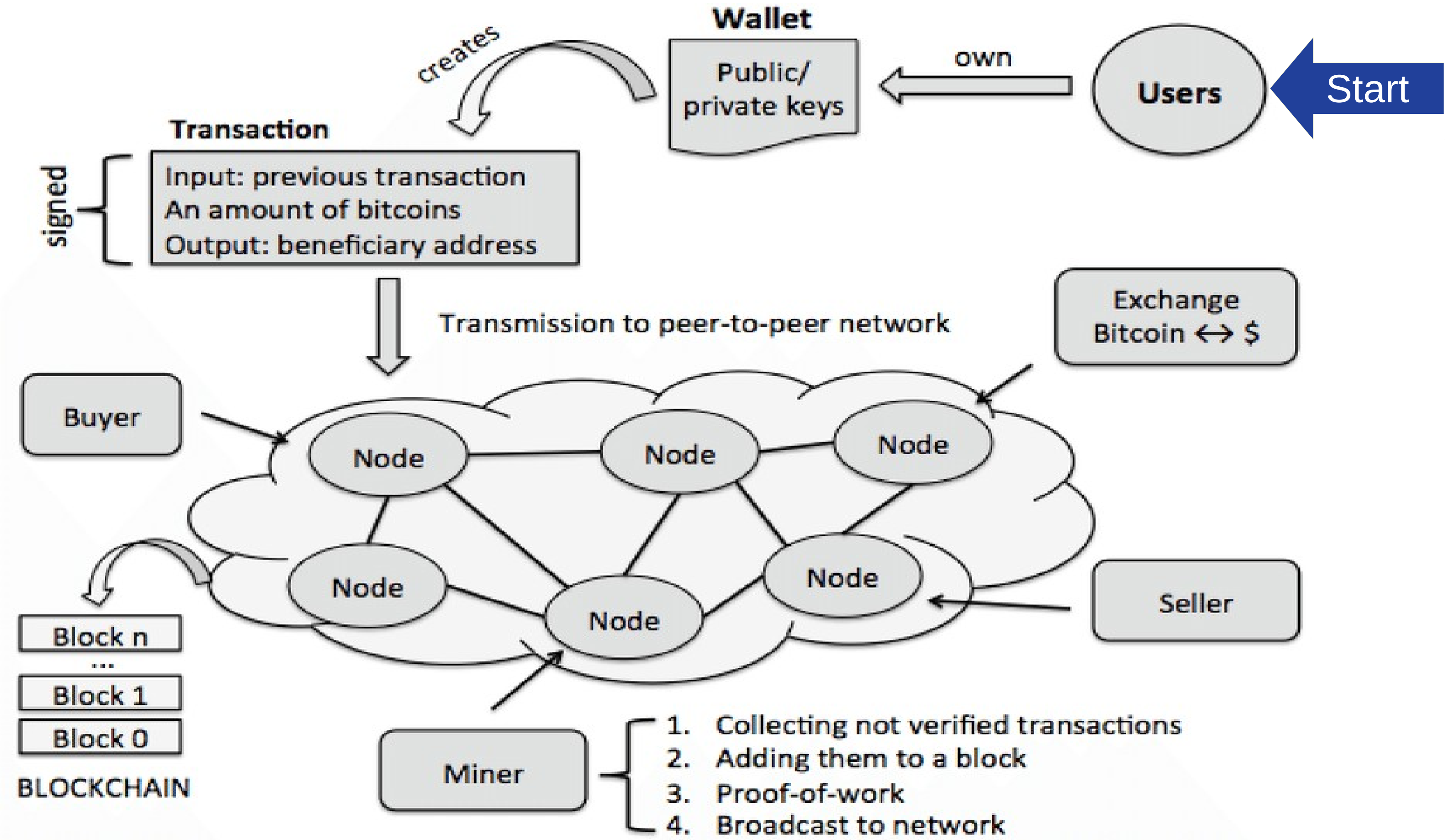
DAPPS

- A decentralized application: DApp, dApp, Dapp, or dapp
- **A computer application** that runs on a distributed computing system.
- It uses distributed ledger technologies (DLT) in various applications such as
  - DeFi - Decentralized Finance
  - **Finance:** Uniswap (Ethereum) – A decentralized exchange (DEX) that enables users to swap cryptocurrencies without intermediaries.
  - **Social Media**: Steemit(Hive blockchain) – A decentralized blogging platform where users earn rewards in cryptocurrency for content creation.
  - **Health care:** Medicalchain (Hyperledger) – A blockchain-based platform that enables secure storage and sharing of electronic health records (EHRs).
  - Many more

Other Applications of BCT

- Financial market
- Media and entertainment
- Energy trading
- Prediction markets
- Retail chains
- Loyalty rewards systems (helps businesses to meet target and give incentives to the right customers)
- Insurance
- Logistics and supply chains(transporting goods to customers)
- Medical records
- Government and military applications

Summary


Wallet
creates
Public/
private keys
own
Users
Start
Transaction
signed
Input: previous transaction
An amount of bitcoins
Output: beneficiary address
Transmission to peer-to-peer network
Exchange
Bitcoin ↔ $
Buyer
Node
Node
Node
Node
Node
Node
Seller
Block n
...
Block 1
Block 0
BLOCKCHAIN
Miner
1. Collecting not verified transactions
2. Adding them to a block
3. Proof-of-work
4. Broadcast to network


BTC
USD
574.65
579.60
0.8828
0.8832

Wallet

- A crypto-currency wallet digitally stores a user’s public and private keys
- Also programmatically helps in sending and receiving digital currency
- **Examples:**
  - **MetaMask** – A popular browser extension and mobile wallet for Ethereum and ERC-20 tokens.
  - **Trust Wallet** – A mobile wallet that supports multiple blockchains.
  - **Exodus** – A multi-asset desktop and mobile wallet with a user-friendly interface.
  - **Coinbase Wallet** – A self-custodial wallet separate from Coinbase Exchange.
  - **Electrum** – A lightweight Bitcoin wallet with strong security features.

Threats & Challenges

- **Double Spending:** One can play with cryptocurrencies trying to spend the same money twice in quick succession.
- While the transaction is evaluated by miners, and takes some time to get confirmed.
- **But before confirmation**, person may send the same amount again to another one.
- So there are 2 unconfirmed transactions in pool called double spending
- In order to avoid this:
  - BCT keeps a timestamp of each transaction

Threats & Challenges

- **Denial of Service Attack:** is a malicious attack on BlockChain by **people who send fack blocks huge in length** to BCT n/w.
- It takes **enormous amount of time for miners** to evaluate the block that reduces the transaction rate of adding blocks to the n/w.
- It happens due to:
  - BCT is public and anyone can access
  - **There is no maximum limit to size of the blocks**
- Some have created a Private / Permissioned BCT where only authorized can access the BCT

Threats & Challenges

- Currently, Miners take
  - Bitcoin (Proof of Work): almost **10 minutes** to mine a bitcoin block which is much more.
  - Ethereum (Proof of Work/Transitioning to Proof of Stake): 13-15 seconds
  - Litecoin (Proof of Work): a new block is mined every 2.5 minutes
  - Cardano (Proof of Stake): 20 seconds
  - Solana (Proof of History): 400 millisecond

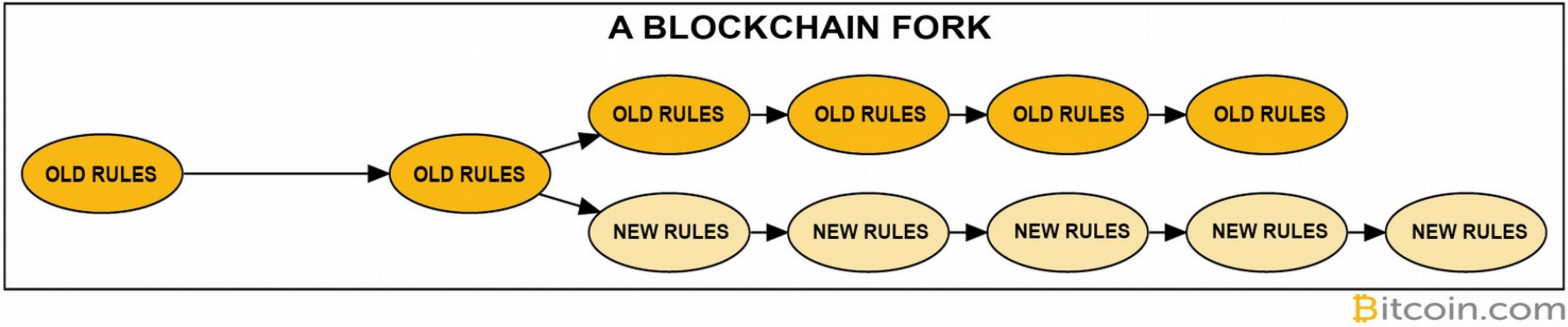
Segwit

- Some Blockchainers applied a new technique called segwit (segregated Witness).
- **Solution bitcoin proposed** : Segwit **impose a restriction on block size** limiting in to only **1 MB**.
- <u>Signature part of the block called “**witness**” would move to trailing part of the block.</u>
- It makes mining much easier and faster due to small size of blocks
- This solution also implemented in crypto-currencies like:
  - Groestlcoin, Litecoin, DigiByte and Vertcoin
- **Segwit2X**: In november 2017, Segwit2X activated:
  - to increase the block size to 2 MB

Threats & Challenges

- **51% Attack:** every framework has a design flaw
- The transaction is confirmed and added to BCT only if a minimum percentage of all nodes accept it
- For bitcoin originally the % was 51%
- In a public BCT **if a miner has 51% hashrate,** then they can manipulate the ledger
- **Ripple uses 80% consensus model.**

Fork in BCT(Soft Fork & Hard Fork)



Fork in BCT(Soft Fork & Hard Fork)

- **Fork means** blocks of different version of BCT
- It occurs when the network's rules or protocols are updated.
- Forks can be categorized into **hard forks and soft forks**
  - based on their impact on **backward compatibility and network consensus.**
- Transactions are added to block and the new block is finally added to the BCT n/w

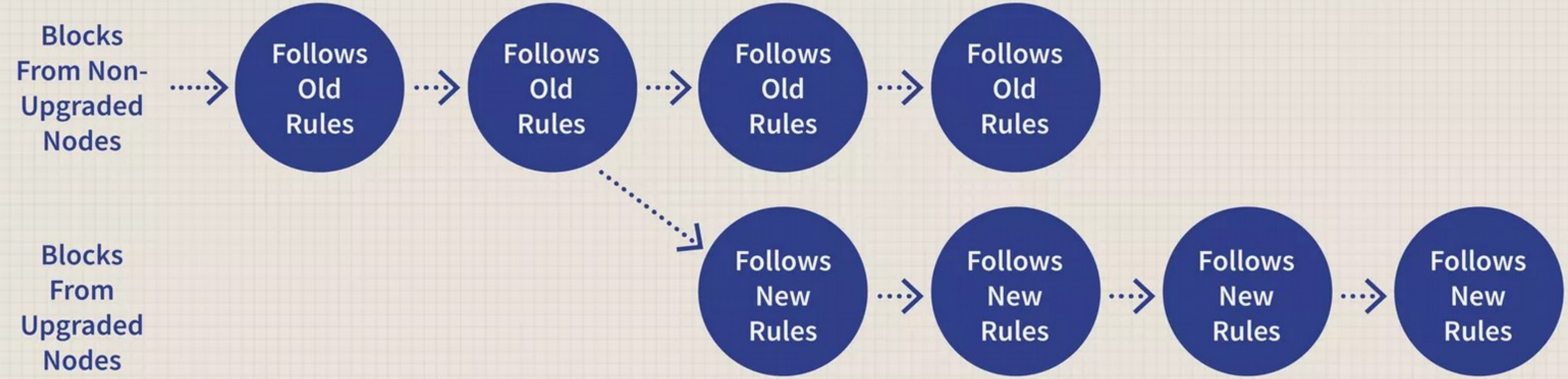
Fork in BCT(Soft Fork & Hard Fork)

- Combined effort of **miners and a distributed consensus to validate** the everyone in n/w using **same version of BCT**
- As people are using different version of BCT, In such case a temporary or accidental fork is created.
- If we use **Ethereum as BCT framework, chances are more for fork**
- However soon is sorted out by getting rid of the faulty blocks. It is called a **soft fork.**

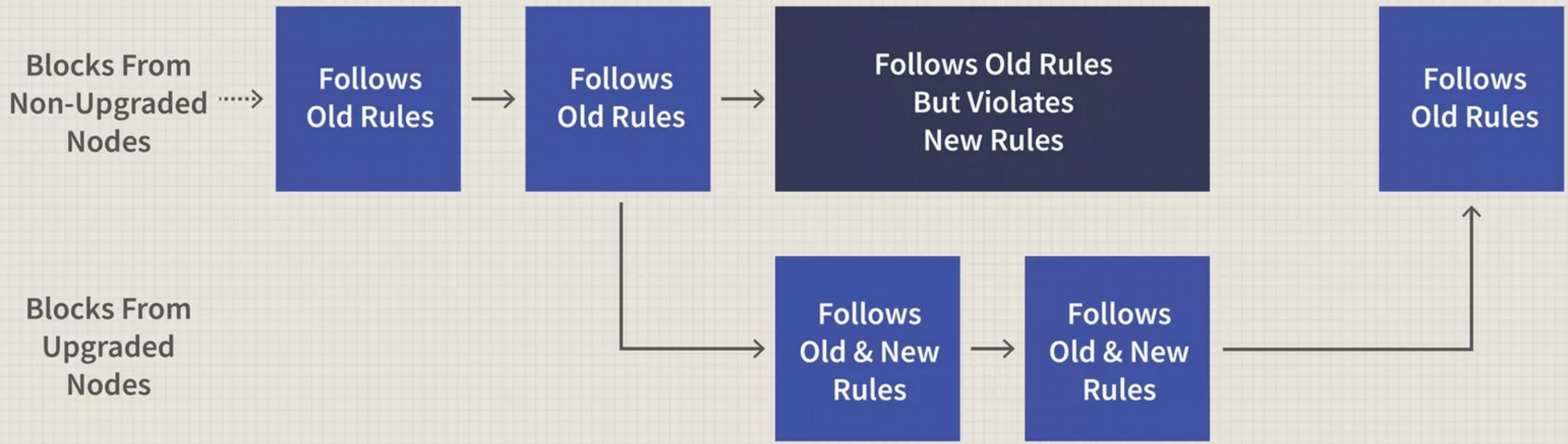
Hard Fork & Soft Fork

- From time to time there is a need in the s/w to change or upgrade
- So <u>two different versions of BCT are created</u> sharing the same origin
- It is called hard fork
- Its a Radical change, new blockchain, not backward-compatible

- A soft fork is a **change to the software protocol**
- **When old nodes will recognize the new blocks as valid**, a soft fork is backwards-compatible.
- Minor change, no new blockchain



A Hard Fork: Non-Upgraded Nodes Reject The New Rules, Diverging The Chain



A Soft Fork: blocks violating new rules are made stale by the upgraded mining majority

Feature	Hard Fork	Soft Fork
Definition	A major change that makes previous versions incompatible with the new version.	A minor upgrade that remains compatible with previous versions.
Backward Compatibility	Not backward-compatible (nodes must upgrade to continue participating).	Backward-compatible (older nodes can still validate transactions).
Consensus Requirement	Requires majority of nodes and miners to upgrade.	Can be implemented even if only a majority of miners adopt it.
Network Split	Causes a permanent split, creating two separate blockchains (old and new).	Does not cause a split, as older nodes still recognize the updated chain.
Example	Bitcoin Cash (BCH) split from Bitcoin (BTC) in 2017 due to block size increase.	SegWit (Segregated Witness) update in Bitcoin, which improved scalability without splitting the chain.
Impact on Transactions	Transactions on the new chain are not recognized by the old chain.	Transactions remain valid on both old and new nodes.
Risk Factor	Higher risk of network division and reduced security.	Lower risk, as old nodes remain functional.

Property / Assets

- Any type of property or asset can be registered on blockchain.
  - physical or digital
  - Laptops, mobile phones, diamonds, Automobiles
  - real estate, E-registrations, digital files,etc.

- From one person to another: maintain the transaction log, and check validity or ownerships.

Other Financial Applications

- cross-border payments
- Share trading
- loyalty and rewards system
- Know Your Customer (KYC) among banks, etc.
- Initial Coin Offering (ICO) is one of the most trending use cases:

  (just like IPOs : as a means to increase the amount of available financing to the company.)

  – ICO (small portion of cryptocurrency / BabyCoin) is the best way of crowdsourcing today by using cryptocurrency as digital assets.
  – **A coin in an ICO can be thought of as a digital stock in an enterprise, which is very easy to buy and trade.**